

VERITRADE: A VERIFIABLE END-TO-END HARNESS FOR DIGITAL TRADE REGULATION

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ABSTRACT

Regulatory indices are compiled by hand: an analyst is given a jurisdiction and a legal question, searches official portals, and records which article answers it. Automating this is usually framed as retrieval over a corpus already assembled. That framing omits three problems — generated citations hallucinate, a correct answer is hard to verify without the source it came from, and the law changes underneath both.

We present **VeriTrade**, an end-to-end harness for **digital trade regulation**. Given a jurisdiction, a topic, and *no source URLs*, it discovers statutes on official portals, acquires them against bot defences and encrypted catalogues, recovers text from scans, segments instruments into articles, retrieves candidates with a script-aware hybrid ranker, maps each to a codified legal test with a model that may never author a citation, and routes by confidence to human review. It covers six Asia-Pacific economies in three writing systems and stores every instrument with a version identity, so rechecking an answer is a query rather than a re-crawl.

We contribute the system, an open implementation that runs on small models, and a reproducible benchmark for verifiable retrieval. On it, one tokenisation decision removes two economies without raising anything. Under a monolingual tokeniser two of the six economies are wholly unreachable, and only tokenisation and native vocabulary together recover them, to 0.395 and 0.250. Every number in this paper is re-derived from run records at compile time.

1 INTRODUCTION

The Regional Digital Trade Integration Index scores an economy’s digital-trade regulation across twelve pillars resolving to sixty-one legal questions (United Nations ESCAP, 2024). Each is answered not with a judgement but with an instrument, an article within it, and that article quoted verbatim. Coding one economy today is a person reading statutes.

Automating this looks like retrieval, and the field has good benchmarks for it: COLIEE (Goebel et al., 2024), LexGLUE (Chalkidis et al., 2022), LegalBench (Guha et al., 2023), and BSARD (Louis & Spanakis, 2022), which poses exactly our question shape. In every one the corpus is handed to the system. That is the right design for measuring a model, and it is why none of them can see the three failures that define the real task.

Citations hallucinate — an audit of commercial legal-research tools sold on their freedom from hallucination found fabricated or misattributed authority in between roughly one in six and one in three responses (Magesh et al., 2024; Dahl et al., 2024), and in law a fabricated authority is worse than none because it is actionable. **Answers are hard to verify**, since checking a legal claim means returning to a source a system assembling evidence transiently cannot offer (Liu et al., 2023). And **the law moves**: instruments are amended, repealed and cut across by neighbours, so a correct answer becomes silently wrong. We treat the first as architectural rather than statistical — the model grades evidence and never writes it, and the one field it may propose is discarded unless it occurs literally in the retrieved snippet — and answer the other two by storing what was read, so rechecking is a query rather than a re-crawl (Section 4.7).

Contributions.

1. **The first end-to-end harness for legal evidence retrieval.** Seven stages from a bare jurisdiction-plus-topic input to cited, confidence-scored indicator rows, with no seed URLs at any point (Section 4). The domain specialisation is digital trade regulation, but nothing in the pipeline is specific to it: a different body of law needs a codified test per indicator, a reference set, and an entry point (Section 4.8). To our knowledge no prior published system combines autonomous source discovery with extraction and indicator mapping in one evaluated pipeline; we state that as a conjunction and engage the closest prior work in Section 2.
2. **Verifiability as an architectural property, not a metric.** The model grades evidence and never writes it: law text, article numbers and URLs are carried from extraction, and the one field the model may propose is discarded unless it occurs literally in the retrieved snippet (Section 4.5). Every row carries the verbatim source text, all four confidence signals, and the portal version it was read from, so a reviewer can reject a mapping directly rather than trusting a score (Sections 4.6, 4.7).
3. **An open implementation that runs on small models.** Recognition and generation are both behind provider interfaces, and the pipeline is engine-agnostic by construction. In a grader bake-off a 24B open-weights model outscored three larger hosted alternatives, and a full run costs cents (Section 4.8).
4. **A reproducible benchmark for verifiable retrieval.** Offline, deterministic, no network and no model: six economies, three scripts, two binary treatments, five seeded draws. It measures *reachability* — whether a query can see a cited instrument at all — and the recall-versus-budget curve above it (Section 5.3). Its numbers, tables and charts are regenerated from records on every build.

2 RELATED WORK

Benchmarks assume the corpus. COLIEE distributes a fixed statute set (Goebel et al., 2024); LexGLUE and LegalBench evaluate over supplied text (Chalkidis et al., 2022; Guha et al., 2023); BSARD pairs questions with 22,600 pre-segmented articles (Louis & Spanakis, 2022). A benchmark whose corpus is given cannot expose a discovery failure, having already paid the cost that dominates the task. A 2026 survey of legal LLM agents (Liu et al., 2026) describes no system that locates sources without seed URLs: its framing is that agents delegate authority to legal databases, which assumes discovery away.

Acquisition and regulatory pipelines. Pile of Law collects US legal material from an enumerated source list (Henderson et al., 2022). Closest to us is LOCUS (Peskoff et al., 2026) — browser automation against anti-bot defences, OCR, segmentation, per-segment classification, four of our seven stages, but over a *known* vendor set, one country, one language, and with no calibration or review routing. REGREACT extracts typed compliance criteria (Ali et al., 2026), ComplianceNLP monitors regulatory change over a curated graph of 12,847 provisions (Guo et al., 2026), and a multi-jurisdictional system spans 68 jurisdictions and 242 documents (Ford et al., 2026); all take their documents as input, which is the boundary this paper moves. BLAD assembles a low-resource statutory corpus (Sakhawat, 2026) and Moran et al. (2026) ask whether regulatory auditing generalises beyond English — both also assume acquisition.

Retrieval and documents. We fuse BM25 (Robertson & Zaragoza, 2009), a dense bi-encoder (Karpukhin et al., 2020) and a cross-encoder (Nogueira et al., 2020) by convex combination rather than reciprocal rank fusion (Cormack et al., 2009), following Bruch et al. (2023). Mr. TyDi and MIRACL evaluate multilingual retrieval *per language* on per-language indexes (Zhang et al., 2021; 2023); the interference in Section 5.3 arises only when scripts share one index. Ogundepo et al. (2022) show character *n*-grams substitute for language-specific tokenizers; we ask instead what a system *reports* with the wrong one. OCR error degrades downstream NLP (van Strien et al., 2020); Aumiller et al. (2021) segment by topical coherence where we recover formal article boundaries; Kamath et al. (2020) show softmax confidence fails under domain shift, load-bearing across six jurisdictions.

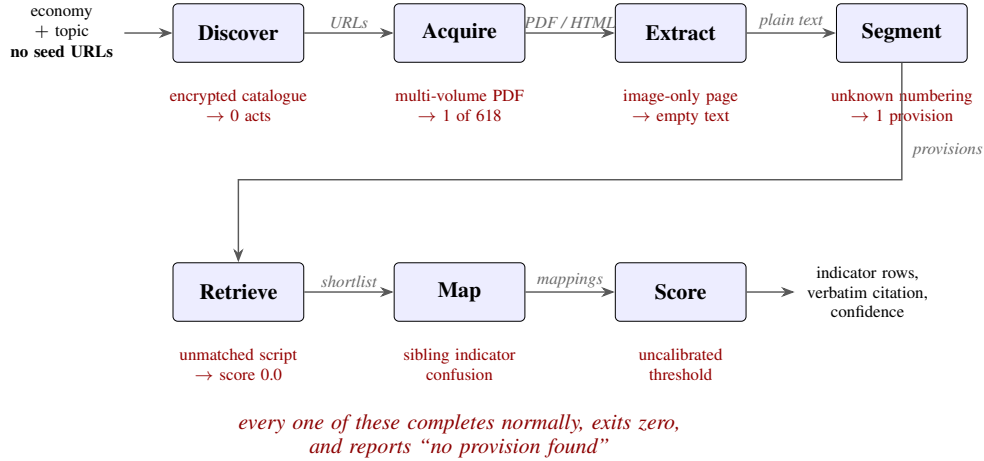


Figure 1: The seven stages, and the silent failure each can produce. Every one of these was observed in development; each completes normally, exits zero, and reports “no provision found” — which is also the correct output for a jurisdiction that genuinely has no such law.

3 PROBLEM FORMULATION

Given a jurisdiction j and indicators $I = \{i_1, \dots, i_k\}$, each carrying a codified legal test, the system must return tuples $\langle \text{instrument, article, verbatim snippet, source URL, confidence} \rangle$ such that the snippet occurs character-for-character in the document at that URL and the article’s operative rule satisfies i ’s test. Where nothing satisfies i the system must say so and name where it looked: a blank is indistinguishable from a failure to search. Three constraints separate this from a retrieval benchmark — **no seed URLs** (the system is told which portals exist and nothing else; no document URL or instrument name may be embedded), **scanned sources count**, and **the answer may not be precomputed**, since evaluation is a live run.

The difficulty concentrates in four places, each returning a confident empty answer when handled naively: statutes sit behind portal software written for human readers, so search surfaces commentary more readily than law; portals resist automated collection; an instrument is not an answer, so segmentation must make the citation checkable; and neighbouring indicators share their vocabulary — a prohibition on transferring data abroad and a permission to transfer it subject to safeguards are written in almost the same words, and retrieval correctly ranks both highly.

4 METHOD

Figure 1 gives the pipeline. We describe each stage and the design decision that the multi-jurisdiction, multi-script setting forced.

4.1 DISCOVERY WITHOUT SEED URLs

The system is configured with one domain per economy and no document URLs. Queries are constructed from the indicators themselves, and split by sub-type: short law-title fragments, and descriptive obligation phrases. The distinction is operational — Australia’s register API matches against the Act title only, so obligation phrases fired at it can never return a hit and merely spend the query budget. Per-indicator terms are interleaved round-robin, so that truncating the query list starves no indicator.

Portals then need portal-specific mechanisms, which we summarise here and detail in Appendix A. Malaysia serves its catalogue as an AES-GCM payload whose key the page publishes for its own JavaScript; we decrypt exactly as the page does. Singapore’s browse listing ignores its pagination parameter entirely, so the index is enumerated by unioning four sort windows. Australia exposes an OData register. India publishes one record per *section* through a DSpace API, so provision

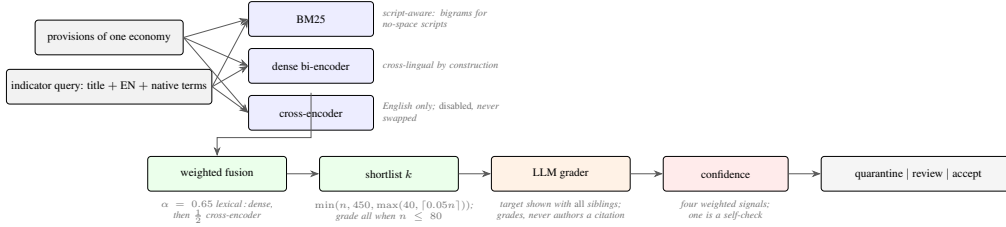


Figure 2: The retrieval stage. Retrieval is indicator-conditioned: the ranker runs once per indicator over the economy’s whole provision set.

boundaries are the publisher’s own. Mongolia injects its body client-side, so the document is reached through the export endpoint its own toolbar calls.

4.2 ACQUISITION

Fetching escalates from a browser-impersonating HTTP client to a stealth browser only when a page proves to be an unrendered application shell. Bodies are content-addressed by SHA-256, so two URLs serving identical bytes occupy one file, and conditional requests avoid refetching.

Two decisions are worth stating. First, `robots.txt` is enforced with unknown treated as denial: a server that cannot tell us its rules has not granted permission. Second, where a portal states that its documents are not to be downloaded — China’s national database returns an explicit `download: 0` permission block — we stop rather than engineer around it, and use a different official portal that publishes the same national laws as ordinary pages. We record this because it is a design constraint with a coverage cost, not a limitation we discovered late.

4.3 RECOGNITION AND SEGMENTATION

Documents are triaged per page, not per file, because a file can carry a text layer on some pages and images on others. Pages with a usable text layer are read directly; the rest are rendered at 300 dpi and passed to OCR. Recognition language is selected per economy from a registry that records, for each engine, whether a model for that script exists at all — and an engine with no model is not invoked. This matters because an engine can only emit characters in its output dictionary: an audit of the shipped recognition dictionaries found the Cyrillic model missing four Mongolian letters and sixteen of Kazakh’s, and the Latin model missing all forty-five precomposed Vietnamese tone forms. A missing letter is not a low score; it is an impossibility that arrives as fluent text.

Segmentation recovers article boundaries per drafting tradition. Singapore and Malaysia number sections; Australia typesets them bold with no keyword; China uses an article marker that PDF extraction routinely splits with spaces; Mongolia numbers articles in laws but uses a decimal clause style in ministerial circulars. Snippets are preserved verbatim: page and heading sentinels are stripped from the snippet but retained in the source text, so a provision spanning a page break still verifies exactly.

4.4 HYBRID RETRIEVAL

Retrieval (Figure 2) scores every provision against every indicator. For provision i and indicator q the fused score is

$$s_i = \frac{1}{2} \left[\max \left(0, \alpha \frac{b_i}{\max_j b_j} + (1 - \alpha) \frac{\cos_i + 1}{2} + \beta_i - \pi_i \right) \right] + \frac{1}{2} \sigma(c_i),$$

with $\alpha = 0.65$, b the BM25 score max-normalised over the corpus, $\cos$ the cosine of a multilingual bi-encoder mapped to $[0, 1]$, σ the logistic, and c the cross-encoder logit, defined as zero outside the rerank pool.

Three design choices carry the multi-script setting.

Tokenisation is script-aware, and the ASCII path is invariant. A run in a script without word spacing becomes overlapping character bigrams; spaced scripts tokenise as words; accented Latin shares a character class with ASCII rather than forming its own. The last point is not cosmetic: giving diacritics their own class shreds Vietnamese into one-letter fragments that match any document, returning plausible non-zero scores computed from noise. Critically, pure-ASCII input tokenises bit-identically to the original English-only rule — pinned by regression test — which is what licenses transferring parameters swept on English corpora to the multilingual system.

Lexical and dense stages see different text. The BM25 corpus includes the law name for document-level discrimination through IDF; the embedded text excludes it, because the law name otherwise dominates the embedding and makes every section of a data-protection act look like a framework hit. Native-language query terms are added to the lexical query only: the bi-encoder is cross-lingual by construction, so an English question already reaches foreign text, while BM25 without native terms scores a flat zero.

The reranker is disabled rather than substituted. The shipped cross-encoder is English-only, and its score is fused at the same weight as BM25 — so an off-language reranker degrades the result rather than failing to help. Selection is keyed on *language*, not script: Vietnamese and Indonesian are Latin-script and equally out of the English model’s depth. Where no model covers the language the stage is switched off entirely: two good signals beat three when the third is arbitrary.

Shortlist size is $k = \min(n, 450, \max(40, \lceil 0.05n \rceil))$, except that a corpus of at most 80 provisions is graded in full, which makes retrieval a signal rather than a gate — our declared mitigation for economies whose retrieval parameters are unmeasured.

4.5 INDICATOR MAPPING

Each candidate is graded independently, and the prompt shows the target indicator *together with all of its siblings*, each with its own legal test. This forces the explicit choice between adjacent indicators that retrieval cannot make. The model returns structured fields — operative rule, whether the target is satisfied, a better sibling if the target is a mislabel, a graded legal-match score, scope, and optionally a subsection — and the contract is explicitly symmetric between over-assignment and miss.

Two constraints define the anti-hallucination boundary. Non-English snippets are reasoned over as written and never translated, because the verbatim column *is* the statute’s text and a translation would make the citation false; only the model’s English rationale is generated. And the sole field the model may contribute to the citation, a subsection reference, is kept only if it is well-formed *and* occurs literally in the snippet. Everything else in the exported row is carried from extraction.

4.6 CONFIDENCE AND ROUTING

Confidence is a weighted sum of four signals: the grader’s legal-match judgement (0.40), the fused retrieval score (0.25), snippet grounding (0.20), and scope alignment (0.15). Grounding is a self-check — whether the quoted snippet is present in the source document — and is the only signal computed without a model. Two hard caps then apply: a scope-mismatched mapping on the comprehensive-framework indicator is capped at 0.55, and a snippet sharing no concept vocabulary with the pillar is capped at 0.45. Rows route to auto-accept, human review, or quarantine. Indicators with no evidence receive an explicit “no provision found” row naming the portal searched; a blank is never emitted.

These six constants are hand-set. Fitting them needs thousands of labelled provision–question pairs including hard negatives, and the published answer key gives only final answers for three economies; generating negatives ourselves would be grading our own work. The response is to surface all four signals and the verbatim snippet on every row, so a reviewer can reject a mapping directly rather than trust an uncalibrated number.

4.7 FRESHNESS: THE LAW MOVES

An answer is only as current as the text behind it, so the unit of storage is a *law version* rather than a document. A law’s identity is its economy and landing URL; a version’s identity is that law plus the portal’s own version signal — a compilation date in Australia, a reprint date in Malaysia. Where a portal exposes no such signal the content hash *is* the version key, which never lies about whether the text changed. Storing a new version supersedes the previous one in the same transaction, so a retrieval pass sees each provision once rather than once per historical edition.

This makes revalidation cheap enough to be routine. Checking whether a published answer is stale means asking the portal for the version signal of the ten to thirty instruments that answer actually cites, not re-crawling a jurisdiction. Each is classified current, stale, or unverifiable — and a portal that does not respond is unverifiable rather than current, because silence is not confirmation. The interference case is handled at grading time rather than storage time: the model is shown the target indicator together with all of its siblings, so an instrument that touches two adjacent obligations is scored against both rather than being assigned to whichever was asked first (Section 4.5).

4.8 PORTABILITY AND MODEL REQUIREMENTS

Recognition and generation both sit behind provider interfaces, and the pipeline selects among them per economy rather than globally: recognition language is chosen from a registry that records, per engine, whether a model for that script exists at all, and an engine with no model is not invoked. The consequence is that adding an engine is a registry entry, and adding a jurisdiction is a configuration entry naming its portal and adapter.

The generation stage is deliberately undemanding. It performs one bounded judgement per candidate — does this provision’s operative rule satisfy this legal test, or does a sibling fit better — against a fixed schema, with no long-context reasoning and no tool use. In a bake-off on 58 hand-read cases using the shipped prompt, a 24B open-weights model reached F1 0.903 against 0.812, 0.800 and 0.786 for three larger hosted alternatives, the decisive difference being recall (0.688 to 0.875) at the same price. Only first place is firm — two models moved by about 0.05 across two runs — but the useful claim is weaker anyway: the architecture does not require a frontier model, and a run costs cents.

Porting to another body of law needs four things, none of them harness changes: a codified test per indicator, a reference set from a source authoritative in that domain, an entry point, and native query vocabulary per language — which Section 5.3 shows is not optional in any multilingual jurisdiction.

4.9 THE PROVENANCE HARNESS

The pipeline’s rule for law — evidence is carried, never generated — is applied one level up to the paper. Experiments write one record per matrix point carrying a content hash of its configuration and inputs; metrics collapse records into cells, aggregating seed axes exactly once so no value reaches a figure without its spread; and a checker walks this document, resolves every cited value back to the records behind it, and fails the build on anything it cannot ground. A condition matching two cells is an error rather than a pick-the-first; a single-seed cell reports its spread as undefined rather than zero; a failed run is recorded as failed rather than scored zero.

5 EXPERIMENTS

5.1 SYSTEM COVERAGE

We ran the pipeline end to end across six economies. **These are single runs against live portals, on the dates shown.** They are readings, not rates: nothing has been run twice on different days, and portals change. We report them because they establish that the pipeline completes across three writing systems, and because the provisions-per-document column locates where it does not yet work.

Appendix D gives the per-economy figures. The diagnostic column is provisions per document: India’s 423 documents yield 938 provisions because most are one- or two-page circulars whose

numbering the segmenter does not yet recognise, which is a segmentation gap rather than a discovery one.

Of thirty-seven instruments the panel cites, twenty-seven are in our catalogue. The other ten are regulator guidance, licence conditions and industry codes — not statutes, and not published where statutes are. That is a discovery gap, not a retrieval one: no reranking reaches a document never enumerated.

5.2 RETRIEVAL PARAMETERS ARE MEASURED, NOT CHOSEN

Every parameter in Section 4.4 was swept against the expert panel’s own answer key on a stratified corpus of 383 laws and 36,590 provisions. The headline: raising the shortlist from 40 to 300 candidates moved provision recall from 15/18 to 18/18, and measured end-to-end on production code the parameter change moved recall from 11/18 to 17/18 — Australia going from 1 of 5 cited provisions retrieved to 5 of 5, with the shortlist getting *cleaner* as it grew. Both fused signals are load-bearing: $\alpha = 0.65$ gives 18/18 while pure BM25 or pure dense each give 16/18, and removing the cross-encoder costs a target while using it alone costs three. Two tempting ideas measured *worse* and are recorded in Appendix E so they are not re-proposed: a law-level prefilter is uniformly harmful at every budget, and per-law slot reservation degenerates into breadth without depth.

5.3 AN ECONOMY CAN DISAPPEAR WITHOUT AN ERROR

The experiments above are single-configuration measurements against a live system. This one is controlled, reproducible offline, and re-derived from records at build time. We cross six economies with two binary treatments — tokenisation (monolingual vs. script-aware) and query vocabulary (English vs. plus native statutory terms) — over 5 seeded draws of a distractor pool, indexing all economies’ instruments together in one pooled index of 2113 documents. Instruments are rendered as their own portal serves them; Mongolia’s are recovered into Cyrillic by matching the numeric instrument identifier inside the source URL, a language-independent join.

We report *reachability*: the share of the panel’s cited instruments that receive any non-zero lexical score at all. A zero here is not a ranking that went badly; it is a document the query cannot see.

Table 1: Reachability by economy and condition. Latin-script economies are unaffected by either treatment, by construction. The two non-Latin economies are unreachable under three of four conditions, and no configuration raises or logs.

Economy	Script	English terms		+ native terms	
		Monolingual	Script-aware	Monolingual	Script-aware
Singapore	Latin	0.6364	0.6364	0.6364	0.6364
Australia	Latin	0.5263	0.5263	0.5263	0.5263
Malaysia	Latin	0.8000	0.8000	0.8000	0.8000
India	Latin	0.7959	0.7959	0.7959	0.7959
China	Han	0.0000	0.0000	0.0000	0.3947
Mongolia	Cyrillic	0.0833	0.0833	0.0833	0.2500

Under the monolingual tokeniser the indicator vocabulary reaches 0.000 of China’s cited instruments and 0.083 of Mongolia’s; making tokenisation script-aware leaves China at 0.000 and Mongolia at 0.083, and only adding native statutory vocabulary raises them to 0.395 and 0.250.

The structure of Table 1 is the finding. Each fix is necessary and neither is sufficient: China stays at 0.0000 once its text is tokenisable, because an English query and a Chinese statute share no tokens whatever the tokeniser does, and at 0.0000 with Chinese phrases under the monolingual rule, because those phrases tokenise to nothing on the way in. The operational point is the mode rather than the magnitude: lexical scoring carries most of the fused score, so such an economy contributes a uniform zero and the pipeline terminates normally reporting no provision found — a legitimate and common answer in a regulatory index.

5.4 WIDENING THE TOKENISER IS NOT FREE

Widening the tokeniser is not free for the languages that already worked: with the ASCII path byte-identical, Australia’s recall@10 moves by -0.105 and Malaysia’s by +0.040, because the two share one index and therefore one set of collection statistics.

Under the monolingual rule the Cyrillic distractors tokenise to nothing: they occupy the index but can never score. Making the tokeniser script-aware admits them, which changes the document-frequency and length-normalisation statistics BM25 shares across the collection (Robertson & Zaragoza, 2009). Australia’s recall@10 falls from 0.2632 to 0.1579 although not one byte of Australian tokenisation changed, while Malaysia rises from 0.3029 to 0.3429. Reachability, a per-document property, is unchanged for every Latin economy; only rank-sensitive metrics move. A team measuring only reachability would have concluded the change was free.

Reachability is a property of the vocabulary and the tokeniser rather than of the corpus sample: over 5 seeded draws of the distractor pool its spread is 0.0000, while the rank-sensitive MRR over the same draws spreads 0.0005.

Reachability does not depend on which distractors were drawn, which is what one should expect of a property of the query vocabulary and the tokeniser, and is worth confirming rather than assuming. The rank-sensitive metrics do move with the draw, which is why the seed is an axis of the grid and every cell carries its spread.

5.5 ONE SHORTLIST BUDGET DOES NOT FIT SIX JURISDICTIONS

The shortlist size is a cost axis, not a free parameter: the grader is billed per candidate, so a configuration that reaches every target by grading everything has answered nothing useful. Figure 3 in Appendix B plots recall against budget under both treatments, with each economy’s reachability drawn as a ceiling; the curves have visibly different shapes.

A single shortlist budget is wrong for every jurisdiction at once. Going from five candidates to fifty lifts Malaysia’s recall from 0.143 to 0.760 and India’s from 0.122 to 0.755, but China saturates at twenty (0.342 there, 0.342 at fifty) and Mongolia gains +0.083 across the whole range – so the extra grading cost buys a great deal in one economy and nothing in another.

A global top- k tuned on one jurisdiction is therefore the wrong budget everywhere else, and where the ceiling binds, buying more candidates cannot help: the missing instruments are not ranked low, they are invisible.

Reachability bounds recall, as it should: at the widest budget measured, Singapore’s recall equals its reachability exactly (0.6364 against 0.6364) and no economy exceeds its own ceiling. A provision the query cannot see is not recovered by grading more candidates.

Reachability is defined as an upper bound on recall and behaves as one at every budget measured — the internal-consistency check the benchmark exists to support, and the reason we report it as the headline metric.

5.6 RECOGNITION QUALITY

On the one bundled scanned instrument with a human-checked transcript, RapidOCR reaches 1.11% character error at about 4 s/page against a 5% requirement. A third engine initially measured 0.00%, and that measurement was an artifact: extracting nothing from an image-only PDF, it fell back to reading the ground-truth transcript shipped beside the file and compared itself against the answer key. Its true error is 95.56% (Appendix C). A benchmark that hands an engine its own reference is not a weak measurement but a measurement of nothing — and it reported a perfect score.

6 LIMITATIONS

Confidence is uncalibrated: four weights and two thresholds are hand-set, ordering rows by reliability without being probabilities. **There is no external baseline:** 18/18 target recall is good relative to our own previous configuration and nothing else. The scorer we substituted caught one of its

own errors — Mongolia initially scored 0 of 9 because the key names instruments in English while the portal serves Mongolian. **Coverage is bounded by discovery, not ranking:** ten of thirty-seven cited instruments are outside the catalogue because they are not statutes, and closing that needs a separately enumerable index per instrument type, most of which do not exist. **Most of the framework is unscored:** nine of sixty-one indicators have codified tests; the other fifty-two run but have never been checked for whether their definitions discriminate. **System-level numbers are single runs against live portals;** only Section 5.3 is reproducible offline. **The controlled experiment measures the lexical floor** — instrument titles and reachability rather than provision text and mapping F1 — because provision-level corpora for three economies do not exist and a live crawl is neither reproducible nor free. We report the layer we can measure and decline to report the one we cannot.

7 CONCLUSION

A legal-evidence pipeline that cannot be audited is one whose failures are invisible, and the multi-lingual ones are the most invisible of all: they present as the output a correct system produces on a jurisdiction with no such law. We built a harness that discovers its own sources, never lets a model author a citation, stores what it read so an answer can be rechecked, and refuses to publish a number it cannot trace to a run — and pointing it at its own retrieval layer found two economies silently absent.

REPRODUCIBILITY STATEMENT

The controlled experiment of Section 5.3 runs offline with no network, no credentials and no language model: both of its inputs — the expert panel’s reference set and the harvested title catalogue — are files in the repository, so its records can be re-derived by anyone who clones it. The full grid is 5 seeds over the conditions of Table 1, and the code under test is imported from the shipped system rather than reimplemented, because a benchmark that measures a copy of a tokeniser measures the copy.

Every number in this paper that appears inside a verified region is a macro resolved at compile time from those records, through the aggregation described in Section 4.9; the document does not build if one stops resolving. Generated tables and charts carry the hash of the metrics file they came from and are re-checked against it at build time. Appendices G and the cell listing give the per-cell means, spreads and sample sizes behind every figure.

The system-level runs of Section 5.1 are *not* reproducible in this sense and we do not present them as though they were: they depend on live government portals whose contents change, and each is a single measurement on the date shown.

ETHICS STATEMENT

This system retrieves public law from official government portals, and the manner of retrieval raises two issues we treat as design constraints rather than implementation details.

Collection. Crawling is rate-limited per host and honours `robots.txt`, with an unreachable directive treated as denial rather than permission — a server that cannot state its rules has not granted any. Where a portal declares that its documents are not to be downloaded we stop and use a different official source that publishes the same instruments, accepting the coverage cost (Section 4.2). One narrow, named exception relaxes certificate validation for two hosts serving public statutes with expired certificates; no credential is ever sent, and every fetched body is hashed so a substituted document is detectable after the fact.

Use. The output feeds a regulatory index that informs policy, so a wrong citation is not a benign error. This motivates the architecture rather than sitting beside it: the model may not author a citation, every row carries the verbatim source text and all four confidence signals, and rows below threshold are routed to human review rather than published. We state explicitly in Section 4.6 that the confidence score is uncalibrated and therefore orders rows by reliability without being a probability — presenting it as one would invite exactly the misplaced trust the system is built to avoid.

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A PORTAL MECHANISMS

Each portal required a different mechanism, and in every case the naive approach returned a plausible empty result rather than an error. Dates are the days the behaviour was verified against the live site.

Malaysia — an encrypted catalogue. The legislation portal answers its catalogue endpoint with `{"encrypted": true, "data": "..."}.` The payload is AES-256-GCM laid out as a 12-byte IV, a 16-byte tag, then ciphertext, and the key is a 64-hex-character constant the page publishes in its own source so that its client-side JavaScript can decrypt it. We do exactly what that script does. This is a transport wrapper rather than an access control — the plaintext is the same public catalogue of Malaysian legislation — but a client that does not implement it reads *zero* Acts and raises nothing. Two catalogues are enumerated, principal and amendment, because the principal one carries dated reprints into which a later amendment has not yet been folded.

Singapore — a pagination parameter that is ignored. The browse listing accepts a page parameter and disregards it: pages one, two and three return byte-identical HTML at every page size, with or without the AJAX header (verified 2026-08-01). The index is therefore not walkable. What the portal does honour is sort order, so asking for the same list ascending and then descending returns its first and last windows, and their union is the whole index whenever the total is at most twice the

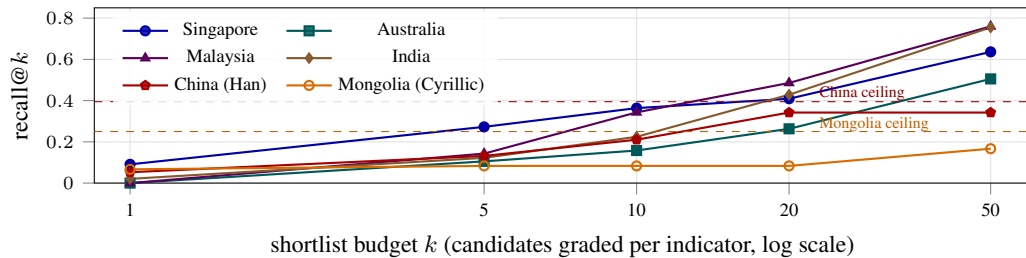


Figure 3: Recall against shortlist budget, with reachability ceilings for the two non-Latin economies. Every coordinate is resolved from run records at compile time. The curves have different shapes: extra budget is worth a great deal in some jurisdictions and nothing in others.

page size. Four windows over two sort keys cover the 524 current Acts. They do *not* cover the 5,843 subsidiary instruments, and the shortfall is logged as incomplete rather than passed over in silence. The same portal throttles a burst by answering HTTP 202 with an empty body rather than 429, so a naive client reads success and an empty page.

Australia — multi-volume compilations. The register exposes an OData API whose filter matches against the Act *title* only, so descriptive obligation phrases can never hit and are suppressed for this portal. Large Acts are published as multiple volumes, and for those the single-file PDF URL returns 404 — the volume number must be appended. Before this was handled the 404 was silent: the fetch failed, the pipeline fell back to the JavaScript landing page, and the Act contributed one junk provision instead of roughly eight hundred. It struck exactly the largest Acts, which is where retention and government-access provisions live. Volumes must also be extracted and segmented *separately* before merging: concatenating them first makes the second volume’s opening body look like a table of contents, which deleted an entire data-retention Part.

India — one record per section. India Code exposes a DSpace API that publishes each *section* as its own item, so provision boundaries are the publisher’s rather than ours and no OCR or segmentation heuristic is involved. Two constraints are non-obvious. The API silently caps a page at 100 items however many are requested, while reporting a page count computed as though it had returned the larger number — so a client that trusts it loses most of a long Act with no error. And queries must be quoted phrases: the unquoted form of a cross-border data query returns the Commercial Courts Act, a highways rule and a police manual, with the actual data-protection statute absent.

Mongolia — a body that is not in the page. The detail page injects its content client-side; the roughly twelve thousand Cyrillic characters in the served shell are navigation, a modal and a sector index. The document is reached through the export endpoint the page’s own toolbar calls, which returns Word-flavoured HTML — so no OCR and no binary parse. Titles are matched per-word rather than as phrases, because Mongolian word order differs from the answer key’s and the portal’s own filenames contain missing spaces; per-word matching also carries agglutination in the right direction, since a stem reaches its declined forms. One detail decided whether the economy worked at all: converting block-level elements to newlines *before* stripping tags. Without it the law arrives as 34,196 characters of correct Mongolian on a single line, from which a line-anchored article pattern finds nothing — and the run completes, extracts, maps and reports no provision found.

China — where we stop. The national law database’s own API returns a permission block containing `download: 0`, and the document path resolves through a signed reader on an internal host. That is the operator stating the documents are not to be downloaded, so we do not engineer around it. The cyberspace administration’s portal publishes the same national laws as ordinary server-rendered pages, and that is the lane used. The cost is real: the provincial tier remains reachable but is the wrong tier for national indicators.

B RECALL AGAINST SHORTLIST BUDGET

C RECOGNITION BENCHMARK

On the one bundled scanned instrument with a human-checked transcript, RapidOCR reaches 1.11% character error at about 4 s/page and PaddleOCR 0.00% at 585 s/page, against a 5% requirement. We report a third number because it is more instructive: MarkItDown initially measured 0.00% error, and the measurement was an artifact — extracting nothing from an image-only PDF, it fell back to reading the ground-truth transcript shipped beside the file and compared itself against the answer key. Measured on a copy in a directory without that sidecar, its true error is 95.56%. The fix was to exclude sidecar-reading engines from measurement and run the benchmark on an isolated copy. We record it because a benchmark that hands an engine its own reference is not a weak measurement; it is a measurement of nothing, and it reported a perfect score.

Measured on the bundled scanned notice: RapidOCR 1.11% character error at 4.07 s/page; PaddleOCR 0.00% at 584.56 s/page, a $143\times$ penalty for an engine of equivalent accuracy; MarkItDown 95.56% once measured on an isolated copy. The fix was to exclude sidecar-reading engines from measurement and to run the benchmark in a directory containing no reference transcript.

D END-TO-END COVERAGE

Table 2: End-to-end single runs. Provisions per document is the diagnostic column: India’s 423 documents yield 938 provisions because most are one- or two-page circulars whose numbering the segmenter does not yet recognise.

Economy	Pillar	Documents	Provisions	Prov./doc.	Rows	Wall clock
Singapore	7	22	5,237	238	107	13 min
Malaysia	6+7	25	5,483	219	46	27 min
Australia	6+7	18	2,196	122	29	11 min
China	7	18	559	31	51	8 min
Mongolia	7	18	272	15	40	5 min
India	7	423	938	2	44	7 min

E RETRIEVAL PARAMETER SWEEP

Parameters were swept against the panel’s own answer key on a stratified corpus of 383 laws and 36,590 provisions, scoring whether the specific provision the panel named reaches the shortlist. Raising the shortlist from 40 to 300 candidates moved provision recall from 15/18 to 18/18. Fusion ablations at the chosen budget: $\alpha = 0.65$ gives 18/18, $\alpha = 0.50$ gives 17/18, and pure BM25 or pure dense each give 16/18 — so both signals are load-bearing. Removing the cross-encoder costs a target; using it alone costs three.

Two negative results are worth recording because both are tempting. A law-level prefilter is *uniformly worse* at every budget: a law whose title is about something else still holds the one provision that answers the indicator, and a law-level gate discards it before any provision is considered. And reserving shortlist slots per law degenerates on a multi-hundred-law corpus into breadth-without-depth. Measured end-to-end on production code, the parameter change moved provision recall from 11/18 to 17/18, with Australia going from 1 of 5 cited provisions retrieved to 5 of 5 — and the shortlist got *cleaner* as it got larger.

F WORKED SEGMENTATION AND RETRIEVAL EXAMPLES

Three instruments, one per script, walked through the same code the controlled experiment measures. The aggregate result of Section 5.3 says an economy is unreachable; these show why, token by token. The Singaporean case is the control — Latin script throughout, so no condition changes anything.

Native titles are reproduced as images. This is not a stylistic choice: the TeX installation used for this build has no CJK package and no Cyrillic encoding, and transliteration is not available to us, because the pipeline’s own rule is that statutory text is carried and never rewritten. What appears below is the statute’s own text rather than a romanisation of it. Non-Latin tokens are counted rather than printed for the same reason, and counted separately from Latin ones so that a mixed-script query shows which half is doing the work.

F.1 CN — P6-I2 (HAN SCRIPT)

Instrument. Personal Information Protection Law of the PRC. The panel’s own answer for local storage, and the article the shipped grading prompt uses as its worked example.

As the portal publishes it:

中华人民共和国个人信息保护法

Condition	Query tokens	Title tokens	Overlap
Monolingual, English terms	local, storage, requirements, stored,... (82)	<i>none</i>	0
Script-aware, English terms	local, storage, requirements, stored,... (82)	13 non-Latin	0
Monolingual, + native terms	local, storage, requirements, stored,... (82)	<i>none</i>	0
Script-aware, + native terms	local, storage, requirements, stored, ... (82) + 33 non-Latin	13 non-Latin	6

F.2 MN — P7-I1 (CYRILLIC SCRIPT)

Instrument. Law on Personal Data Protection. Recovered into its own script by the portal-id join; the panel recorded it under an English name only.

As the portal publishes it:

ХҮНИЙ ХУВИЙН МЭДЭЭЛЭЛХАМГААЛАХ ТУХАЙ

Condition	Query tokens	Title tokens	Overlap
Monolingual, English terms	comprehensive, legal, framework, for,... (59)	<i>none</i>	0
Script-aware, English terms	comprehensive, legal, framework, for,... (59)	4 non-Latin	0
Monolingual, + native terms	comprehensive, legal, framework, for,... (59)	<i>none</i>	0
Script-aware, + native terms	comprehensive, legal, framework, for, ... (59) + 10 non-Latin	4 non-Latin	2

F.3 SG — P7-I1 (LATIN SCRIPT)

Instrument. Personal Data Protection Act 2012. The control. Latin script throughout, so no condition changes anything.

As the portal publishes it:

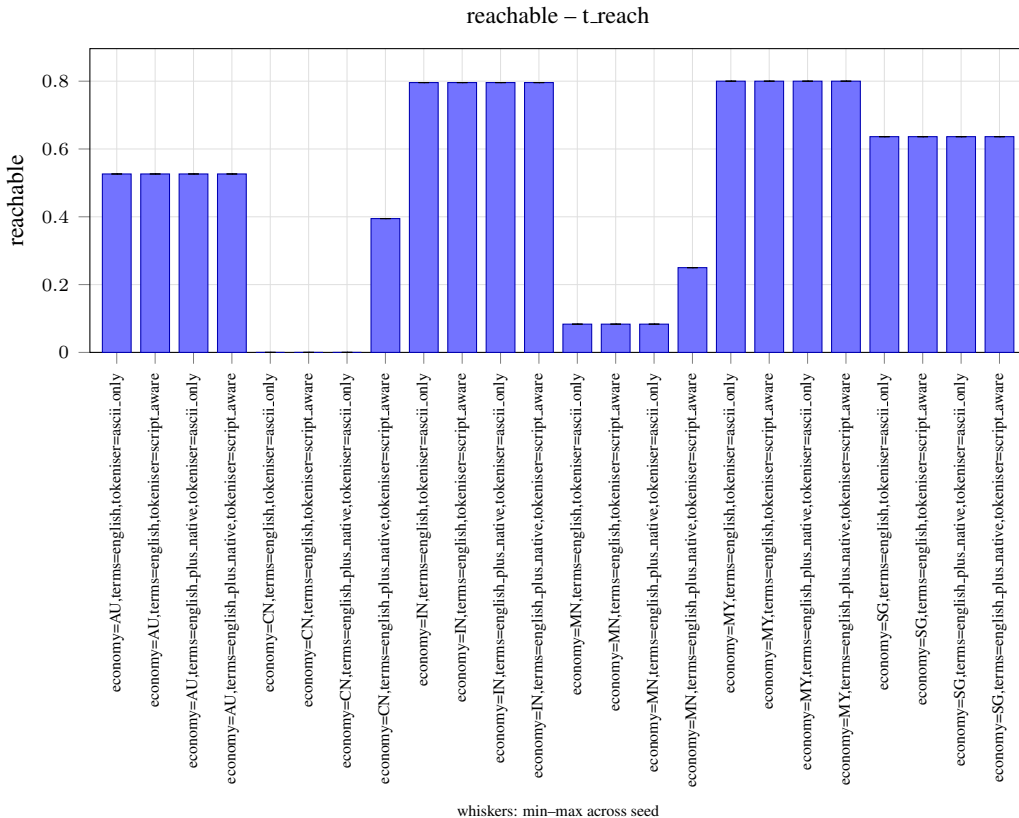
Personal Data Protection Act 2012

Condition		Query tokens	Title tokens	Overlap
Monolingual, terms	English	comprehensive, legal, framework, for,... (59)	personal, data, protection, act, ... (5)	4
Script-aware, terms	English	comprehensive, legal, framework, for,... (59)	personal, data, protection, act, ... (5)	4
Monolingual, terms	+ native	comprehensive, legal, framework, for,... (59)	personal, data, protection, act, ... (5)	4
Script-aware, terms	+ native	comprehensive, legal, framework, for,... (59)	personal, data, protection, act, ... (5)	4

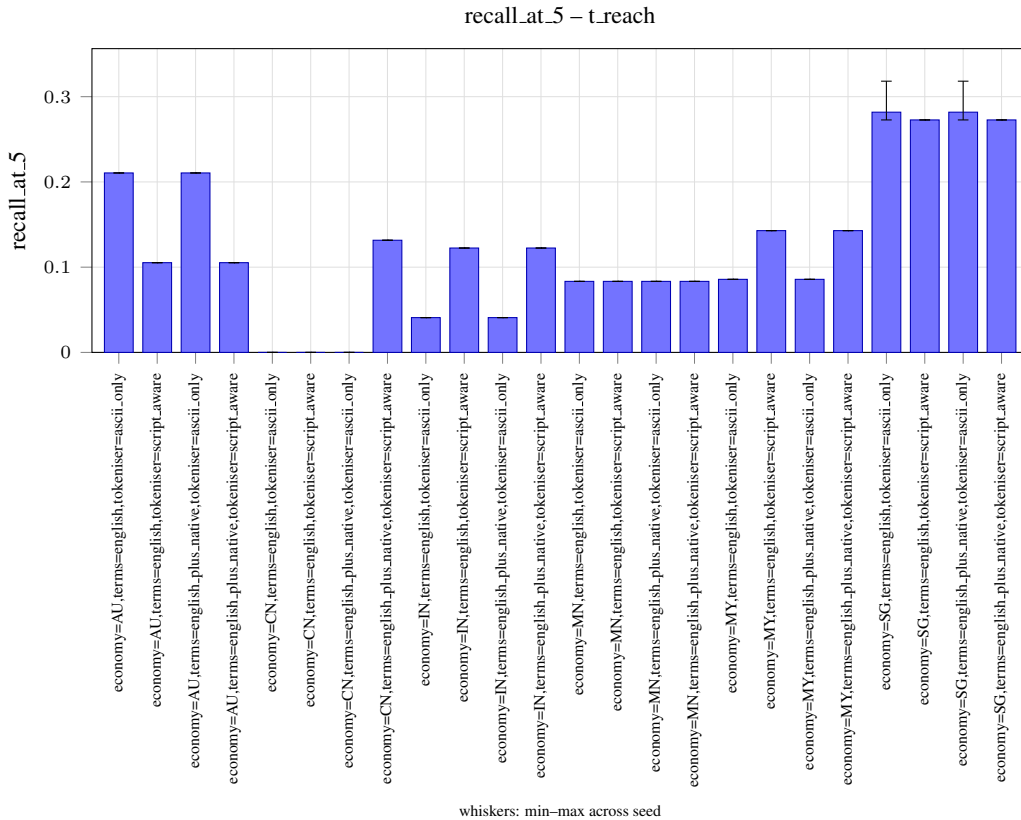
G GENERATED CHARTS

Emitted by the figures stage from the same metrics file as every table above. Bars are cell means; whiskers run min to max across the seed axes. A cell measured at one seed draws no whisker, because a whisker of zero would read as a measured spread of zero.

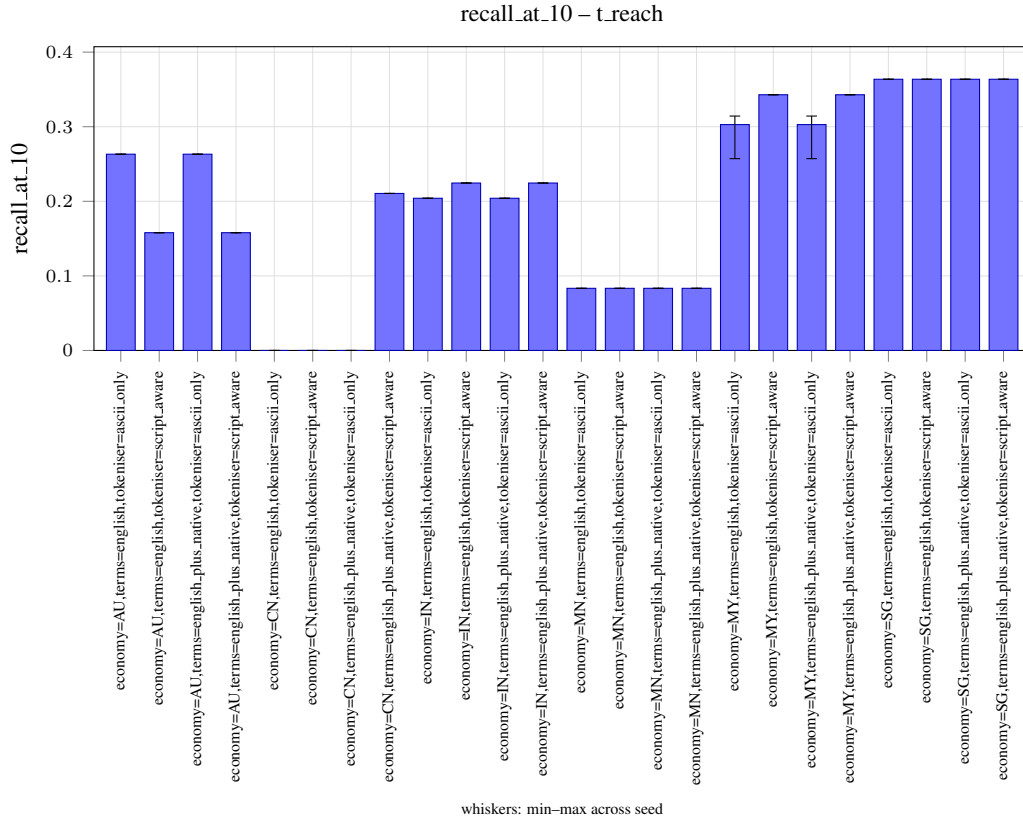
G.1 REACHABILITY



G.2 RECALL@5



G.3 RECALL@10



H FULL CELL LISTING

H.1 REACHABILITY

Condition	Mean	SD	Spread	<i>n</i>	Failed
economy=AU,terms=english,tokeniser=ascii_only	0.5263	0.0000	0.0000	5	0
economy=AU,terms=english,tokeniser=script_aware	0.5263	0.0000	0.0000	5	0
economy=AU,terms=english_plus_native,tokeniser=ascii_only	0.5263	0.0000	0.0000	5	0
economy=AU,terms=english_plus_native,tokeniser=script_aware	0.5263	0.0000	0.0000	5	0
economy=CN,terms=english,tokeniser=ascii_only	0.0000	0.0000	0.0000	5	0
economy=CN,terms=english,tokeniser=script_aware	0.0000	0.0000	0.0000	5	0
economy=CN,terms=english_plus_native,tokeniser=ascii_only	0.0000	0.0000	0.0000	5	0
economy=CN,terms=english_plus_native,tokeniser=script_aware	0.3947	0.0000	0.0000	5	0
economy=IN,terms=english,tokeniser=ascii_only	0.7959	0.0000	0.0000	5	0
economy=IN,terms=english,tokeniser=script_aware	0.7959	0.0000	0.0000	5	0
economy=IN,terms=english_plus_native,tokeniser=ascii_only	0.7959	0.0000	0.0000	5	0
economy=IN,terms=english_plus_native,tokeniser=script_aware	0.7959	0.0000	0.0000	5	0
economy=MN,terms=english,tokeniser=ascii_only	0.0833	0.0000	0.0000	5	0
economy=MN,terms=english,tokeniser=script_aware	0.0833	0.0000	0.0000	5	0
economy=MN,terms=english_plus_native,tokeniser=ascii_only	0.0833	0.0000	0.0000	5	0
economy=MN,terms=english_plus_native,tokeniser=script_aware	0.2500	0.0000	0.0000	5	0
economy=MY,terms=english,tokeniser=ascii_only	0.8000	0.0000	0.0000	5	0
economy=MY,terms=english,tokeniser=script_aware	0.8000	0.0000	0.0000	5	0
economy=MY,terms=english_plus_native,tokeniser=ascii_only	0.8000	0.0000	0.0000	5	0
economy=MY,terms=english_plus_native,tokeniser=script_aware	0.8000	0.0000	0.0000	5	0
economy=SG,terms=english,tokeniser=ascii_only	0.6364	0.0000	0.0000	5	0
economy=SG,terms=english,tokeniser=script_aware	0.6364	0.0000	0.0000	5	0
economy=SG,terms=english_plus_native,tokeniser=ascii_only	0.6364	0.0000	0.0000	5	0
economy=SG,terms=english_plus_native,tokeniser=script_aware	0.6364	0.0000	0.0000	5	0

H.2 RECALL@5

Condition	Mean	SD	Spread	n	Failed
economy=AU,terms=english,tokeniser=ascii_only	0.2105	0.0000	0.0000	5	0
economy=AU,terms=english,tokeniser=script_aware	0.1053	0.0000	0.0000	5	0
economy=AU,terms=english_plus_native,tokeniser=ascii_only	0.2105	0.0000	0.0000	5	0
economy=AU,terms=english_plus_native,tokeniser=script_aware	0.1053	0.0000	0.0000	5	0
economy=CN,terms=english,tokeniser=ascii_only	0.0000	0.0000	0.0000	5	0
economy=CN,terms=english,tokeniser=script_aware	0.0000	0.0000	0.0000	5	0
economy=CN,terms=english_plus_native,tokeniser=ascii_only	0.0000	0.0000	0.0000	5	0
economy=CN,terms=english_plus_native,tokeniser=script_aware	0.1316	0.0000	0.0000	5	0
economy=IN,terms=english,tokeniser=ascii_only	0.0408	0.0000	0.0000	5	0
economy=IN,terms=english,tokeniser=script_aware	0.1224	0.0000	0.0000	5	0
economy=IN,terms=english_plus_native,tokeniser=ascii_only	0.0408	0.0000	0.0000	5	0
economy=IN,terms=english_plus_native,tokeniser=script_aware	0.1224	0.0000	0.0000	5	0
economy=MN,terms=english,tokeniser=ascii_only	0.0833	0.0000	0.0000	5	0
economy=MN,terms=english,tokeniser=script_aware	0.0833	0.0000	0.0000	5	0
economy=MN,terms=english_plus_native,tokeniser=ascii_only	0.0833	0.0000	0.0000	5	0
economy=MN,terms=english_plus_native,tokeniser=script_aware	0.0833	0.0000	0.0000	5	0
economy=MY,terms=english,tokeniser=ascii_only	0.0857	0.0000	0.0000	5	0
economy=MY,terms=english,tokeniser=script_aware	0.1429	0.0000	0.0000	5	0
economy=MY,terms=english_plus_native,tokeniser=ascii_only	0.0857	0.0000	0.0000	5	0
economy=MY,terms=english_plus_native,tokeniser=script_aware	0.1429	0.0000	0.0000	5	0
economy=SG,terms=english,tokeniser=ascii_only	0.2818	0.0203	0.0455	5	0
economy=SG,terms=english,tokeniser=script_aware	0.2727	0.0000	0.0000	5	0
economy=SG,terms=english_plus_native,tokeniser=ascii_only	0.2818	0.0203	0.0455	5	0
economy=SG,terms=english_plus_native,tokeniser=script_aware	0.2727	0.0000	0.0000	5	0

H.3 RECALL@10

Condition	Mean	SD	Spread	n	Failed
economy=AU,terms=english,tokeniser=ascii_only	0.2632	0.0000	0.0000	5	0
economy=AU,terms=english,tokeniser=script_aware	0.1579	0.0000	0.0000	5	0
economy=AU,terms=english_plus_native,tokeniser=ascii_only	0.2632	0.0000	0.0000	5	0
economy=AU,terms=english_plus_native,tokeniser=script_aware	0.1579	0.0000	0.0000	5	0
economy=CN,terms=english,tokeniser=ascii_only	0.0000	0.0000	0.0000	5	0
economy=CN,terms=english,tokeniser=script_aware	0.0000	0.0000	0.0000	5	0
economy=CN,terms=english_plus_native,tokeniser=ascii_only	0.0000	0.0000	0.0000	5	0
economy=CN,terms=english_plus_native,tokeniser=script_aware	0.2105	0.0000	0.0000	5	0
economy=IN,terms=english,tokeniser=ascii_only	0.2041	0.0000	0.0000	5	0
economy=IN,terms=english,tokeniser=script_aware	0.2245	0.0000	0.0000	5	0
economy=IN,terms=english_plus_native,tokeniser=ascii_only	0.2041	0.0000	0.0000	5	0
economy=IN,terms=english_plus_native,tokeniser=script_aware	0.2245	0.0000	0.0000	5	0
economy=MN,terms=english,tokeniser=ascii_only	0.0833	0.0000	0.0000	5	0
economy=MN,terms=english,tokeniser=script_aware	0.0833	0.0000	0.0000	5	0
economy=MN,terms=english_plus_native,tokeniser=ascii_only	0.0833	0.0000	0.0000	5	0
economy=MN,terms=english_plus_native,tokeniser=script_aware	0.0833	0.0000	0.0000	5	0
economy=MY,terms=english,tokeniser=ascii_only	0.3029	0.0256	0.0571	5	0
economy=MY,terms=english,tokeniser=script_aware	0.3429	0.0000	0.0000	5	0
economy=MY,terms=english_plus_native,tokeniser=ascii_only	0.3029	0.0256	0.0571	5	0
economy=MY,terms=english_plus_native,tokeniser=script_aware	0.3429	0.0000	0.0000	5	0
economy=SG,terms=english,tokeniser=ascii_only	0.3636	0.0000	0.0000	5	0
economy=SG,terms=english,tokeniser=script_aware	0.3636	0.0000	0.0000	5	0
economy=SG,terms=english_plus_native,tokeniser=ascii_only	0.3636	0.0000	0.0000	5	0
economy=SG,terms=english_plus_native,tokeniser=script_aware	0.3636	0.0000	0.0000	5	0